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Fourier series approximation of fractal functions

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ABSTRACT

The fractal function is generated as an attractor of iterated function systems. This article examines the Fourier series representation of fractal interpolation functions, including affine and non-affine functions with different vertical scaling factors, as a way of expressing fractal functions explicitly. The given concepts have been illustrated with appropriate examples and graphical approaches.

1. Introduction

In real-life applications, irregular shapes are excluded from Euclidean geometry. In order to overcome this difficulty, Mandelbrot[1] proposed the fractal notion. His research aims to find solutions to complex structural problems in real life. Fractal objects can be constructed using many mathematical techniques, including Iterated Function System (IFS) theory, random fractals, finite subdivision rules, chaotic behavior, and Lindenmayer systems. Hutchinson proposed the theory of fractal sets based on iterated function systems, wherein fixed point theory assumes a pivotal role as an integral constituent of the IFS. Moreover, the concept of fractals has been popularized by Barnsley MF. The framework of fractal set is an invariant set or attractor that prescribes a set of contraction functions defined on a non-empty closed and bounded subset of a complete metric space.[2,3] For fractal objects, this provides a broad foundation for their visual beauty.

The theory of traditional interpolation explains how continuous functions satisfy actual data and how they are transformed. Although traditional interpolation is an effective method of approximation theory, most research has focused on smooth functions. In the case of non-smooth continuous functions on compact sets, the Fractal Interpolation Function (FIF) is a generalization of the traditional interpolation function. The graph of an invariant set of iterated function systems[4] is employed to generate fractal interpolation functions. Over the past thirty years, fractal function theory has developed beyond its mathematical framework and has undergone extensive research into the geometric shapes of fractal functions in various types of spaces.[5] In Ref. 6, the authors have studied the fractal functions are applicable to signal analysis. The FIF theory to regenerate self-similar fractal objects by exploring them at Lorenz chaotic attractor were examined.[7]

Many mathematicians have developed fractal functions such as linear, quadratic, hidden variable, bilinear, coalescence hidden variable, α-fractal functions, Hermite fractal interpolation, multivariate zipper, etc. (for more information Refs. 8–14). The concept of linear recurrent fractal interpolation function was examined for interpolating a data set influenced by Gaussian noise. Further, the flexibility at any intermediate value within this noisy data set was analyzed by exploring the magnitude factor of the probability distribution associated with the fractal function.[15] The zipper fractal functions on a compact domain have employed the conventional spline and cubic interpolation theory. Moreover, the shape-preserving attributes of fractal functions were presented using magnitude factors, and numerical examples were provided to demonstrate these properties.[16] In Ref. 17, they analyzed the fractal interpolation function associated with iterated function system explored through various types of contraction functions.

In this sequence, the α-fractal function with various scaling vectors as a fluctuation of a set of continuous functions on a closed interval in $\mathbb{R}$ was analyzed (for example, Refs. 18, 19). The notion of bivariate alpha FF has been explored across various disciplines, including shape preservation through the magnitude factor of IFS, fundamental theorems, and range determined fractal approximative on a compact domain.[20]

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Numerous mathematicians have been investigated the properties of α–fractal functions, including roughness, differentiability class, fractal dimension, bound estimates, a novel approach using Bernstein polynomials, and the fractal dimension in various spaces such as continuous spaces and function spaces (for more information, see Refs. 21–26)

The theory of Fourier series has mesmerizing connections with numerous disciplines of pure and applied mathematics. The Fourier series on a compact domain have uniform convergence to continuous and bounded functions.[27] Further, the explicit form of the Fourier series becomes a nonsmooth function as the harmonic iterations increase. Fourier series represent periodic signals based on sine and cosine waves. Fourier series are employed in numerous applications, including vibration analysis, electrical engineering, quantum physics, image processing, and signal processing. As a collection of periodic and continuous functions in the Banach space, the author investigated the relationship between fractal functions and standard trigonometric polynomials.[28] The basic properties of generalized Fourier interpolation of fractal functions and their nature of convergence were analyzed.[29] While the notion of the Bernstein fractal function, which corresponds to trigonometric concepts, was examined.[30] According to fractal theory, the trigonometric series plays a crucial role. Numerous suitable families of functions have been employed to evaluate nonsmooth functions utilizing this approach. For instance, fractal rational trigonometric and periodic phenomena of fractal functions have been developed through the fractal operator, and their approximation properties have been analyzed (for more information, see Refs. 31–37).

In Ref. 38, Fourier-Jacobi expansions of fundamental features like bounded variation and uniform convergence have been studied. The Fourier-Jacobi expansion for affine and non-affine fractal functions were presented. Furthermore, they discussed their uniform convergence in $L^2(-1,1)$.[39] An analysis of the fractal shape of the magnetic island in tokamak plasma and their dimensions is carried out by using the radial magnetic field perturbation.[40] In Ref. 41 explored the correlation between state analysis technology and health monitoring signal analysis using multi fractal detrend analysis. The integrated fractal model in three dimensions has been used to study functional linguistics of resonance characteristics based on semantic wave frequency in various kinds of systems.[42] Different types of algorithms have been employed to create three-dimensional animation combinations of fractal structures in Ref. 43 A Fourier series structure for a fractal function with a vertical scale vector has been investigated.[44] Fourier series could represent a relatively complicated fractal interpolation function. Due to the aforementioned research, researchers have not examined the interaction between various fractal functions (affine and non-affine) with variable scaling factors and the Fourier series. Through the Fourier series, the present study explores the explicit structure of different kinds of fractal functions with variable scaling factors. Since nonsmoothness is essential for approximating the complexity of irregular signals, the Fourier series associated with fractal functions offers the closest approach to indicating periodic signals.

This article is organized as follows: Section 2 provides a basic overview. In Section 3, we demonstrate fractal functions using a variable scaling vector derived from the Fourier series, as well as explicit structures for fractal functions derived from the Fourier sine and cosine series. In Section 4, the α-fractal interpolation function is illuminated by calculating function scaling factors based on the Fourier series, and explicit structure interpretations are presented for α-fractal functions based on Fourier sine and cosine series.

2. Preliminaries

Let K be the natural number strictly greater than one. Let $[u_0, u_K]$ and $[u_{s-1}, u_s]$ be the compact interval of $\mathbb{R}$, denoted as J and J_s for $s = 1, 2, \dots, K$ respectively. Let $\{(u_s, v_s) \in J\times\mathbb{R} : s = 0, 1, 2, \dots, K\}$ be the collection of coordinate points such that $u_{s-1} < u_s$ for all $s = 1, 2, \dots, K$. The FIF corresponding with the given data points is constructed as follows. Let l_s be a bicontinuous mapping from J to J_s such that

$$|l_s(u) - l_s(u')| \le r_s|u - u'|,$$

where $\forall u, u' \in J$, r_s is contractivity ratio, which lies between the clopen interval $[0, 1)$ and meets the join-up terms

$$l_s(u_0) = u_{s-1},\ l_s(u_K) = u_s,\ \forall\ s = 1, 2, \dots, K. \tag{1}$$

Let M be a closed and bounded subset of $J\times\mathbb{R}$ and let $\mathbf{f}_s$ be delineated contraction functions with respect to the second variable

$$|\mathbf{f}_s(u, v) - \mathbf{f}_s(u, v')| \le c_s|v - v'|,$$

where $\forall v, v' \in \mathbb{R}, u \in [u_0, u_K]$, and the contraction factor $c_s \in [0, 1)$. Which obeys the join-up terms

$$\mathbf{f}_s(u_0, v_0) = v_{s-1},\ \mathbf{f}_s(u_K, v_K) = v_s,\ \forall\ s = 1, 2, \dots, K. \tag{2}$$

The following system

$$\{M; \mathbf{g}_s : s = 1, 2, \dots, K\}, \tag{3}$$

is referred to as an IFS. Let $\mathbf{g}_s : M \to J_s \times \mathbb{R}, s = 1, \dots, K$ be the finite set of contraction mappings structured as

$$\mathbf{g}_s\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \beta_s & 0 \\ \xi_s & c_s \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} + \begin{pmatrix} \gamma_s \\ \eta_s \end{pmatrix}, \tag{4}$$

where β_s, ξ_s, γ_s, and η_s are belongs to $\mathbb{R}$ and $|c_s| < 1$ is identifying as vertical scaling factor.

Let $\mathcal{K}(M)$ be signified as the hyperspace of all non-void closed and bounded subsets of M and which is complete according to the Hausdorff metric. A self mapping W is delineated on $\mathcal{K}(M)$ by

$$W(A^*) = \bigcup_{s=1}^{K} \mathbf{g}_s(A^*),$$

for any $A^* \in \mathcal{K}(M)$. As W is the contraction on $\mathcal{K}(M)$, therefore according to the IFS theory,[3] W possesses a unique set $G_{\mathbf{h}}$ such that $G_{\mathbf{h}} = W(G_{\mathbf{h}})$, where $G_{\mathbf{h}} := \{(u, \mathbf{h}(u)) : u \in [u_0, u_K]\}$. The function $\mathbf{h}$ is the graph of invariant set of the IFS in Eq. (3) correlation with the data points $\{(u_s, v_s) \in [u_0, u_K]\times\mathbb{R} : s = 0, 1, 2, \dots, K\}$ is said to be the FIF or Fractal Function. Delineate the operator T^* on $C(J)$, by

$$T^*(\mathbf{h}(u)) = \mathbf{f}_s\left(l_s^{-1}(u), \mathbf{h}\left(l_s^{-1}(u)\right)\right),\ u \in J_s,\ s = 1, 2, \dots, K,$$

gratifying $\mathbf{h}(u_0) = u_0$ and $\mathbf{h}(u_K) = u_K$. As the fixed point of T^* is the function $\mathbf{h}$, gratifies the ensuing functional equation

$$\mathbf{h}(u) = \mathbf{f}_s\left(l_s^{-1}(u), \mathbf{h}\left(l_s^{-1}(u)\right)\right),\ u \in J_s,\ s = 1, 2, \dots, K$$

or

$$\mathbf{h}(l_s(u)) = c_s\mathbf{h}(u) + q_s(u),\ u \in J_s,\ s = 1, 2, \dots, K. \tag{5}$$

The kind of comprehensively examined fractal function thus far is delineated by the IFS in Eq. (3) associated with the prescribed data set as follows:

$$l_s(u) = \beta_s u + \gamma_s,\ \mathbf{f}_s(u, v) = c_s v + q_s(u),\ \forall\ s = 1, 2, \dots, K,$$

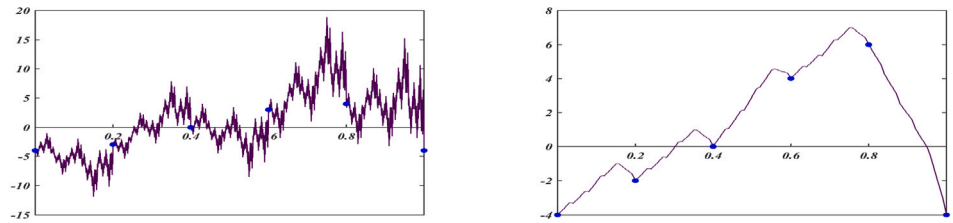
where

$$\beta_s = \frac{u_s - u_{s-1}}{u_K - u_0}, \gamma_s = \frac{u_K u_{s-1} - u_0 u_s}{u_K - u_0},\ \forall\ s = 1, 2, \dots, K, \tag{6}$$

and $c_s = (c_1, c_2, \dots, c_K)$ is the scaling vector and q_s is a Lipschitz function to be employed by

$$q_s(u_0) = v_{s-1} - c_s v_0,\ q_s(u_K) = v_s - c_s v_k.$$

The fractal function with both constant and variable scaling vector is studied in Refs. 7, 10. The kind of FIF showing in (4) by employing the constant scaling coefficient as the variable scaling vector, viz $c_s(u)$ are continuous mappings delineated on J, gratifying $\|c_s\|_\infty = \max\left\{c_s(u) : u \in J\right\} < 1$ for $s = 1, 2, \dots, K$ and q_s is Lipschitz functions



(a) FIF with constant scaling factors (b) FIF with Variable scaling factors

Fig. 1. (a) FIF of the coordinate points on $[0,1]$ with step size $l = 1/5, y = (-4,-2,0,4,6,-4)$ with constant scaling factor $c_1 = -0.4, c_2 = 0.4, c_3 = -0.5, c_4 = 0.7, c_5 = 0.8$. (b) Shows the graph of similar observations with the variable scaling vector $c_1(u) = \sin(u), c_2(u) = -0.3u, c_3(u) = 0.5 - \cos(u), c_4(u) = 0.25u, c_5(u) = 0.3/e^u$.

to be utilized by $q_s(u_0) = v_{s-1} - c_s(u_0)v_0$, $q_s(u_K) = v_s - c_s(u_K)v_K$. The composed FIF with variable scaling vector satisfies the functional equation is

$$\mathbf{h}(l_s(u)) = c_s(u)\mathbf{h}(u) + q_s(u),\ u \in J_s,\ s = 1,2,\dots,K. \tag{7}$$

Example 2.1. Consider the fractal function of $\mathbf{h}$ elapse the coordinate points $\{(0,-4),(1/5,-2),(2/5,0),(3/5,4),(4/5,6),(1,-4)\}$ with constant scaling factor $c_1 = -0.4, c_2 = 0.4, c_3 = -0.5, c_4 = 0.7, c_5 = 0.8$. Then, the IFS determines linear FIF corresponding to the given data points is consisting of the following functions $l_1(u) = 0.2u, l_2(u) = 0.2u+0.2, l_3(u) = 0.2u + 0.4, l_4(u) = 0.2u + 0.6, l_5(u) = 0.2u + 0.8$, which is corresponding to the $\mathbf{f}_s$'s are $\mathbf{f}_1 = -0.4v + 4.4u - 5.6, \mathbf{f}_2 = 0.4v - 0.4u - 0.4, \mathbf{f}_3 = -0.5v + 7u - 2, \mathbf{f}_4 = 0.7v - 2.2u + 6.8, \mathbf{f}_5 = 0.8v - 8.8u + 9.2$. Fig. 1(*a*) represents the graphical illustration of FIF with constant scaling vector, while Fig. 1(*b*) indicates the graph of similar data points illuminates of fractal function with variable scaling factors $c_1(u) = \sin(u), c_2(u) = -0.3u, c_3(u) = 0.5 - \cos(u), c_4(u) = 0.25u, c_5(u) = 0.3/e^u$.

3. Fourier series of FIF with function scaling factor

This section examines the Fourier series representation of fractal function with variable scaling vector on a compact interval $[0,1]$. Further, an example of fractal function with both constant and variable parameter is presented.

In Ref. 27, let $\mathbf{h}(u)$ be the periodic function with period $2k$ is of the form as

$$\mathbf{h}(u) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left(A_n\cos\left(\frac{n\pi u}{k}\right) + B_n\sin\left(\frac{n\pi u}{k}\right)\right), \tag{8}$$

is known as Fourier series, where

$$A_0 = \frac{1}{k}\int_{-k}^{k}\mathbf{h}(u)du, \tag{9}$$

$$A_n = \frac{1}{k}\int_{-k}^{k}\mathbf{h}(u)\cos\left(\frac{n\pi u}{k}\right)du, \quad n = 1,2,\dots, \tag{10}$$

$$B_n = \frac{1}{k}\int_{-k}^{k}\mathbf{h}(u)\sin\left(\frac{n\pi u}{k}\right)du, \quad n = 1,2,\dots, \tag{11}$$

where A_0, A_n and B_n in Eqs. from (9) to (11) are called Fourier coefficients of $\mathbf{h}$, it has a period of $2k$. If $\mathbf{h}(u)$ is an even function then B_n is zero and (8), becomes

$$\mathbf{h}(u) = \frac{A_0}{2} + \sum_{n=1}^{\infty}A_n\cos\left(\frac{n\pi u}{k}\right). \tag{12}$$

Eq. (12) is called the Fourier cosine series where Fourier coefficients are in Eqs. (9) and (10). If $\mathbf{h}(u)$ is an odd function then A_n is zero and (8), becomes

$$\mathbf{h}(u) = \sum_{n=1}^{\infty}B_n\sin\left(\frac{n\pi u}{k}\right), \tag{13}$$

which is known as Fourier sine series, where Fourier coefficient is in Eq. (11).

Theorem 3.1. *If $\mathbf{h} : [0,1] \to \mathbb{R}$ is the FIF composed by the transformations $l_s(u) = \beta_s u + \gamma_s, \mathbf{f}_s(u,v) = c_s(u)v + q_s(u)$, where $q_s(u) = \xi_s u + \eta_s$ and $\|c_s(u)\| < 1$. Then Fourier cosine series of $\mathbf{h}(u)$ is indicated as*

$$\begin{aligned}\mathbf{h}(u) = {} & \frac{1}{K}\sum_{s=1}^{K}\left(\frac{1}{2}\xi_s + \eta_s\right) + \frac{1}{K}\sum_{s=1}^{K}\left(c_s(t)\int_0^1\mathbf{h}(t)dt - \int_0^1\left(c_s'(t)\int_0^1\mathbf{h}(t)dt\right)dt\right)\\ & + \sum_{n=1}^{\infty}\left(\frac{2}{n\pi}\sum_{s=1}^{K}\xi_s\sin\frac{sn\pi}{K} - \frac{4K}{n^2\pi^2}\sin\frac{n\pi}{K}\sum_{s=1}^{K}\xi_s\sin\frac{n\pi(2s-1)}{K} + \frac{2}{K}\sum_{s=1}^{K}\left(c_s(t)\int_0^1\right.\right.\\ & \left.\mathbf{h}(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\left(c_s'(t)\int_0^1\mathbf{h}(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right)\\ & \left. + \frac{4}{n\pi}\sin\frac{n\pi}{2K}\sum_{s=1}^{K}\eta_s\cos\frac{n\pi(2s-1)}{K}\right)\cos(n\pi u).\end{aligned}$$

Proof. Consider the affine transformation

$$l_s(u) = \beta_s u + \gamma_s, \mathbf{f}_s(u,v) = \xi_s u + c_s(u)v + \eta_s \quad s = 1,2,\dots,K.$$

Since $\mathbf{h}$ is a continuous function on $[0,1]$ in (8). If $\mathbf{h}$ is an even function then $\mathbf{h}$ could be indicated by Fourier cosine series. Fourier coefficients A_n can be determined the half range interval $[0,1]$ as follows

$$A_0 = \frac{1}{k}\int_{-k}^{k}\mathbf{h}(u)du = 2\int_0^1\mathbf{h}(u)du = 2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}}\mathbf{h}(u)du.$$

Let the transformation $u = l_s(t)$, where $u = \frac{1}{K}t + \frac{s-1}{K}$ and also $du = \frac{1}{K}dt$, so we have

$$A_0 = 2\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))d\left(\frac{1}{K}t + \frac{s-1}{K}\right) = \frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))dt.$$

Applying the functional equation, we get,

$$A_0 = \frac{1}{K}\sum_{s=1}^{K}(2\xi_s + \eta_s) + \frac{1}{K}\sum_{s=1}^{K}\left(c_s(t)\int_0^1\mathbf{h}(t)dt - \int_0^1\left(c_s'(t)\int_0^1\mathbf{h}(t)\right)dt\right). \tag{14}$$

Now find the Fourier coefficient of A_n for $n = 1,2,\dots$

$$A_n = 2\int_0^1\mathbf{h}(u)\cos(n\pi u)\,du = 2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}}\mathbf{h}(u)\cos(n\pi u)\,du.$$

Again utilizing the transformation and substituting the functional equation, we obtain,

$$\begin{aligned}A_n = {} & \frac{2}{K}\sum_{s=1}^{K}\left(\int_0^1 c_s(t)\mathbf{h}(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right.\\ & \left. + \xi_s\int_0^1 t\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt + \eta_s\int_0^1\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right).\end{aligned}$$

Simplifying the above equation we have,

$$\begin{aligned}A_n = {} & \sum_{s=1}^{K}\left(\frac{2}{n\pi}\xi_s\sin\frac{sn\pi}{K} - \xi_s\frac{4K}{n^2\pi^2}\sin\frac{n\pi}{K}\sin\frac{n\pi(2s-1)}{K}\right.\\ & + \frac{2}{K}\left(c_s(t)\int_0^1\mathbf{h}(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)du - \int_0^1\left(c_s'(t)\int_0^1\mathbf{h}(t)\right.\right.\end{aligned} \tag{15}$$

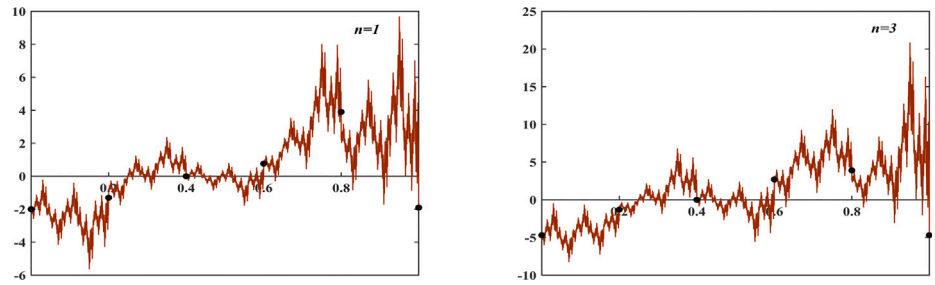


Fig. 2. The fractal function of Fourier cosine series with constant scaling factors are generated by $n = 1, 3$.

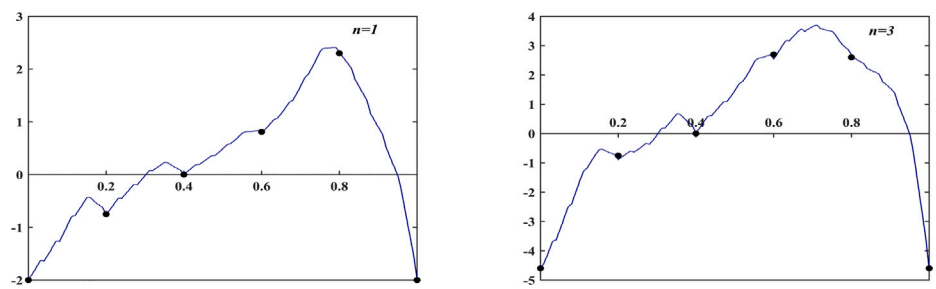


Fig. 3. The FIF constituted by Fourier cosine series for the given data points with function scaling factors are generated by $n = 1, 3$.

$$\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\Bigg)dt\Bigg) + \frac{4}{n\pi}\sin\frac{n\pi}{2K}\eta_s\cos\frac{n\pi(2s-1)}{K}\Bigg).$$

Remark 3.1. The Fourier series constitution of fractal function of variable scaling vector is explained in Theorem 3.1. Xuezai Pan et al. studied Fourier series constitution of fractal function of constant parameters in Ref. 44. As a consequence, the reader can clearly see that Theorem 3.1 is a prolongation of Theorem 1 in Ref. 44. That is, the proof of Theorem 1 in Ref. 44 may be produced if the function scaling factor in Theorem 3.1 is selected as constants in place of function,

$$\begin{aligned}\mathbf{h}(u) = {} & \frac{1}{K}\sum_{s=1}^{K}\left(\frac{1}{2}\xi_s+\eta_s\right) + \frac{1}{K}\sum_{s=1}^{K}c_s\int_0^1\mathbf{h}(t)dt + \sum_{n=1}^{\infty}\Bigg(\frac{2}{n\pi}\sum_{s=1}^{K}\xi_s\sin\frac{sn\pi}{K} \\ & -\frac{4K}{n^2\pi^2}\sin\frac{n\pi}{K}\sum_{s=1}^{K}\xi_s\sin\frac{n\pi(2s-1)}{K} + \frac{2}{K}\sum_{s=1}^{K}c_s\int_0^1\mathbf{h}(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt \\ & +\frac{4}{n\pi}\sin\frac{n\pi}{2K}\sum_{s=1}^{K}\eta_s\cos\frac{n\pi(2s-1)}{K}\Bigg)\cos(n\pi u).\end{aligned}$$

Example 3.1. Consider the data points $\{(0,-4),(1/5,-2),(2/5,0),(3/5,4),$
$(4/5,6),(1,-4)\}$ with constant scaling factor $c_1 = -0.4, c_2 = 0.4, c_3 = -0.5, c_4 = 0.7, c_5 = 0.8$. The graph of fractal interpolation function constituted by the Fourier cosine series generated at $n = 1, 3$ is illustrated in Fig. 2, while the same data points and function scaling factors $c_1(u) = \sin(u), c_2(u) = -0.3u, c_3(u) = 0.5 - \cos(u), c_4(u) = 0.25u, c_5(u) = 0.3/e^u$ is illustrated in Fig. 3.

The fractal function with a variable scaling factor indicated in Fig. 3 is smoother and more elegant than the constant scaling factor offered in Fig. 2, both represented by the Fourier cosine series. Fig. 3 demonstrates that the fractal function roughness is substantially less severe than that seen in Fig. 2 when $n = 1, 3$.

Theorem 3.2. *If* $\mathbf{h}$ *is the FIF delineated on the compact interval* $[0,1]$ *composed by the transformations* $l_s(u) = \beta_s u + \gamma_s, \mathbf{f}_s(u,v) = c_s(u)v + q_s(u)$, *where* $q_s(u) = \xi_s u + \eta_s$ *and* $c_s(u)$ *is variable scaling factor whose range lies between* $(-1,1)$, *then the Fourier sine series of* $\mathbf{h}$ *is given by*

$$\begin{aligned}\mathbf{h}(u) = \sum_{n=1}^{\infty}\Bigg(& -\frac{2}{n\pi}\sum_{s=1}^{K}\xi_s\cos\frac{sn\pi}{K} + \frac{4K}{n^2\pi^2}\sin\frac{n\pi}{K}\sum_{s=1}^{K}\xi_s\cos\frac{n\pi(2s-1)}{K} \\ & +\frac{2}{K}\sum_{s=1}^{K}\Bigg(c_s(t)\int_0^1\mathbf{h}(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\Bigg(c_s'(t)\int_0^1\mathbf{h}(t)\ dt \\ & \sin\left(\frac{n\pi}{K}(t+s-1)\right)\Bigg)\Bigg) + \frac{4}{n\pi}\sin\frac{n\pi}{2K}\sum_{s=1}^{K}\eta_s\sin\frac{n\pi(2s-1)}{K}\Bigg)\sin(n\pi u).\end{aligned}$$

Proof. Consider the affine transformation

$$l_s(u) = \beta_s u + \gamma_s, \mathbf{f}_s(u,v) = c_s(u)v + \xi_s u + \eta_s,\ \ s = 1,2,\ldots,K.$$

Since $\mathbf{h}$ is a continuous mapping on $[0,1]$ in (8). If $\mathbf{h}$ is an odd function then $\mathbf{h}$ is constituted by Fourier sine series. Fourier coefficients $B_n, n = 1,2,\ldots$ is determined the half range interval $[0,1]$ as follows

$$B_n = 2\int_0^1\mathbf{h}(u)\sin(n\pi u)\,du = 2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}}\mathbf{h}(u)\sin(n\pi u)\,du.$$

Let the transformation $u = l_s(t)$, where $u = \frac{1}{K}t + \frac{s-1}{K}$ and also $du = \frac{1}{K}dt$, so we have

$$B_n = 2\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))\sin\left(n\pi l_s(t)\right)\frac{1}{K}dt. \tag{16}$$

Substituting the functional equation in B_n and by using the integration by parts formula to simplify the aforementioned Eqn. (16), then we get

$$\begin{aligned}B_n = {} & -\frac{2}{n\pi}\sum_{s=1}^{K}\xi_s\cos\frac{sn\pi}{K} + \frac{4K}{n^2\pi^2}\sin\frac{n\pi}{K}\sum_{s=1}^{K}\xi_s\cos\frac{n\pi(2s-1)}{K} \\ & +\frac{2}{K}\sum_{s=1}^{K}\Bigg(c_s(t)\int_0^1\mathbf{h}(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\Bigg(c_s'(t)\int_0^1\mathbf{h}(t)\ dt \\ & \sin\left(\frac{n\pi}{K}(t+s-1)\right)\Bigg)\Bigg) + \frac{4}{n\pi}\sin\frac{n\pi}{2K}\sum_{s=1}^{K}\eta_s\sin\frac{n\pi(2s-1)}{K}.\end{aligned} \tag{17}$$

As from the Eq. (17), a complete form of the Fourier sine series of $\mathbf{h}$ in the half range on $[0,1]$ has been mentioned.

Remark 3.2. Theorem 3.2 explains the Fourier sine series construction of FIF with variable scaling vector. Xuezai Pan et al. investigated Fourier sine series formation of FIF with constant scaling vector in Ref. 44. As a result, the reader may observe that Theorem 3.2 is a continuation of Theorem 2 in Ref. 44. That is suppose the function scale

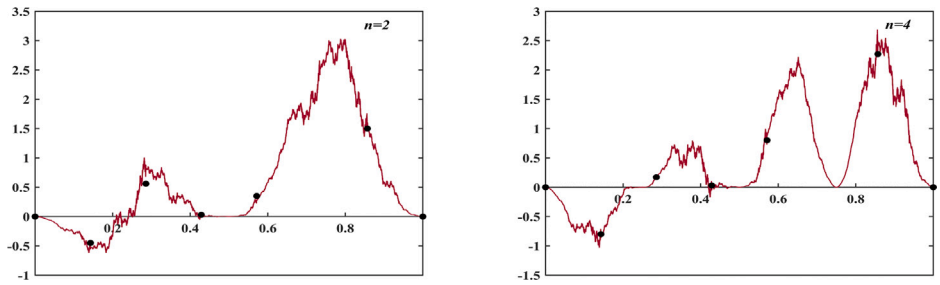


Fig. 4. The fractal function constituted by Fourier sine series of the provided set of data points with constant scaling factors are generated by $n = 2, 4$.

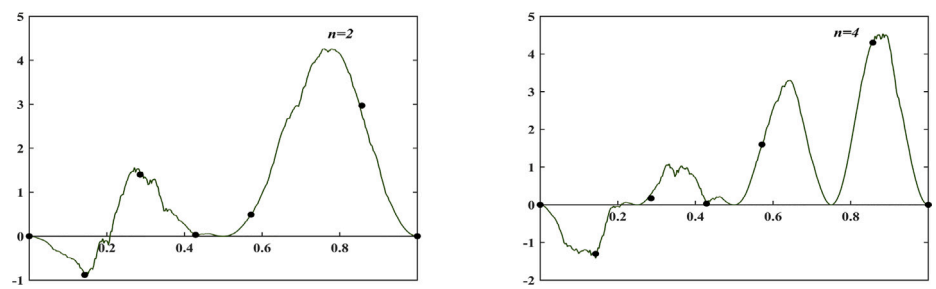


Fig. 5. The fractal function constituted by Fourier sine series of the prescribed data points with function scaling factors are generated by $n = 2, 4$.

factor in Theorem 3.2 are chosen as constants instead of function, the proof of Theorem 3.2 in Ref. 44 is constructed as follows,

$$\mathbf{h}(u) = \sum_{n=1}^{\infty}\Bigg(-\frac{2}{n\pi}\sum_{s=1}^{K}\xi_s\cos\frac{sn\pi}{K} + \frac{4K}{n^2\pi^2}\sin\frac{n\pi}{K}\sum_{s=1}^{K}\xi_s\cos\frac{n\pi(2s-1)}{K} + \frac{2}{K}\sum_{s=1}^{K}c_s\int_0^1\mathbf{h}(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt + \frac{4}{n\pi}\sin\frac{n\pi}{2K}\,dt \sum_{s=1}^{K}\eta_s\sin\frac{n\pi(2s-1)}{K}\Bigg)\sin(n\pi u). \tag{18}$$

Remark 3.3. For an affine function with transformation $u = l_s(t)$, $\mathbf{h}(l_s(t))$ need not be either an even or odd function. The linear fractal function approximates the prescribed set of data points and is defined based on $q_s(u)$ as the affine and non-affine case. Suppose the affine case $q_s(u) = \xi_s u + \eta_s$, this is a linear function so the linear function is neither an even nor odd function.

Example 3.2. Consider the set of data points $u_s = (0, 1/7, 2/7, 3/7, 4/7, 6/7, 1)$ and $v_s = (-5, -3, 3, 0, 5, 9, 7)$, where s varies from 0 to 6 and the constant scaling factors $c_1 = 0.2, c_2 = -0.25, c_3 = 0.3, c_4 = -0.4, c_5 = 0.5, c_6 = -0.6$. The graph of fractal interpolation function constituted by the Fourier sine series generated at $n = 2, 4$ is illustrated in Fig. 4, while the same data points and function scaling factors $c_1(u) = \sin(u), c_2(u) = -0.3u, c_3(u) = 0.5 - \sin(u), c_4(u) = 0.25u, c_5(u) = 0.3/e^u, c_6(u) = 0.5 - \cos(u)$ is illustrated in Fig. 5.

The graph of the fractal function using the Fourier sine series with a variable scaling factor, indicated in Fig. 5, is smoother than the graph with a constant scaling factor, shown in Fig. 4. Fig. 5 reveals that the irregularities of the fractal function are much less severe than those in Fig. 4. Similarly, Figs. 6 and 7 portrays the fractal function of the Fourier series on the closed interval $[0, 1]$. Compared to Figs. 1(a) and 1(b), the fluctuations are expressed similarly, but the ranges are different.

4. Fourier series construction of α-fractal function

The magnitude factors provide to the fractal function an excellence of flexibility and let us to switch its traits. The following part, the function q_s will be delineated through any continuous mappings $\mathbf{h}$ and p_s that map $[0, 1]$ to $\mathbb{R}$, where $\mathbf{h}(0) = p_s(0), \mathbf{h}(1) = p_s(1)$ but $\mathbf{h}$ is not equal to p_s, as

$$q_s(u) = \mathbf{h}(l_s(u)) - c_s(u)p_s(u), \tag{19}$$

which satisfy the joint up conditions $q_s(0) = \mathbf{h}(l_s(0)) - c_s(0)p_s(0), q_s(1) = \mathbf{h}(l_s(1)) - c_s(1)p_s(1)$. This occurrence has presented a generalization of any continuous mapping (for more details[4,18–24]). The α-fractal function constituted of $\mathbf{h}$ is delineated as next after:

Suppose $\mathbf{h}^\alpha$ is the collection of continuous mappings on $[0, 1]$, whose graph is the invariant of IFS defined by (3) and (19) correlated is referred as the α-Fractal Function (namely, α-FIF) according to $\mathbf{h}$ in terms of base functions p_s and the partition so as to $0 = u_{s-1} < u_s = 1$, and is notated as Δ.

$$l_s(u) = \beta_s u + \gamma_s, \tag{20}$$

$$\mathbf{f}_s(u, v) = c_s(u)v + q_s(u), \tag{21}$$

where $q_s(u)$ is aforementioned Eq. (19). As stated in (5), $\mathbf{h}^\alpha$ satisfies the functional equation is

$$\mathbf{h}^\alpha(u) = c_s(u)\left(\mathbf{h}^\alpha(l_s^{-1}(u)) - p_s(l_s^{-1}(u))\right) + \mathbf{h}(u), \ \forall \ u \in [0, 1] \tag{22}$$

(or)

$$\mathbf{h}^\alpha(u) = c_s(u)\left(\mathbf{h}^\alpha - p_s\right)\circ(l_s^{-1}(u)) + \mathbf{h}(u), \ \forall \ u \in [0, 1].$$

Take into account that $\alpha = 0, \mathbf{h}^\alpha$ equal $\mathbf{h}$. Therefore the Eq. (22) may be regarded as an entire collection of functions $\mathbf{h}^\alpha$, with $\mathbf{h}$ as its basis. This approach can be used to identify the fractal equivalents of any continuous mapping.

Example 4.1. Let $\mathbf{h}(u) = 2u^3$ and $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$ be the partition of a compact interval $[0, 1]$. Fig. 8(a) represents the graph of α-Fractal function on $[0, 1]$ with constant scaling vectors are $c_1 = 0.4, c_2 = -0.4, c_3 = 0.5, c_4 = -0.5, c_5 = -0.5$. Let $p_s : [0, 1] \to \mathbb{R}$ be a continuous mapping on $[0, 1]$. Suppose $p(u) = 2u$ gratifies the joint-up provisos $\mathbf{h}(0) = p(0)$ and $\mathbf{h}(1) = p(1)$ but $\mathbf{h}(u) \neq p(u)$. The graphical approach of α-FIF with function scale factors are $c_1(u) = -0.3u/6, c_2(u) = 0.5 - \sin(10u), c_3(u) = -cos(u/15), c_4(u) = (u+1)/5, c_5(u) = 0.5u$, with same base functions in Fig. 8(b).

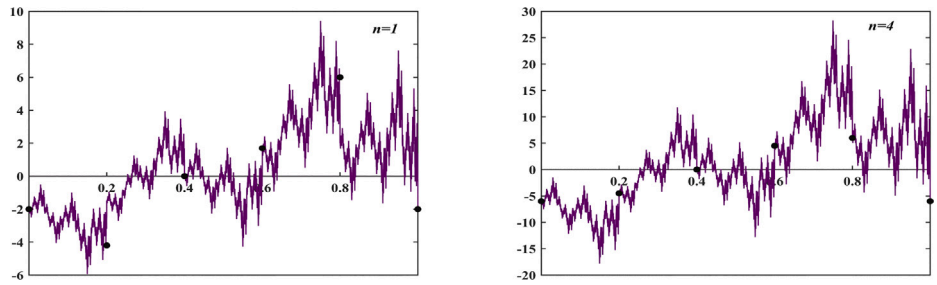


Fig. 6. The fractal function constituted by Fourier series of the set of data points $\{(0,-4),(1/5,-3),(2/5,0),(3/5,3),(4/5,4),(1,-4)\}$ with constant scaling factor $c_1=-0.4, c_2=0.4, c_3=-0.5, c_4=0.7, c_5=0.8$ are generated by $n=1,4$.

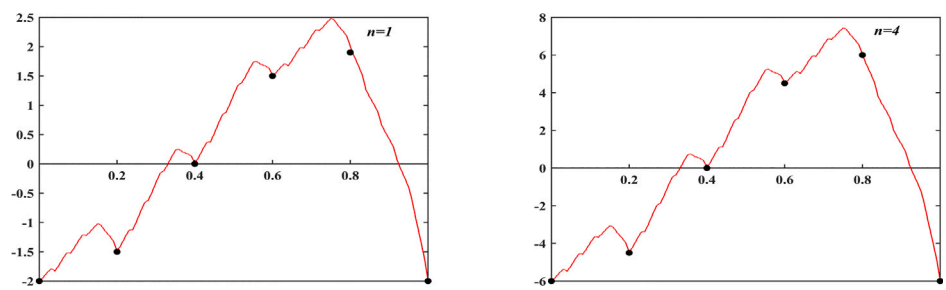
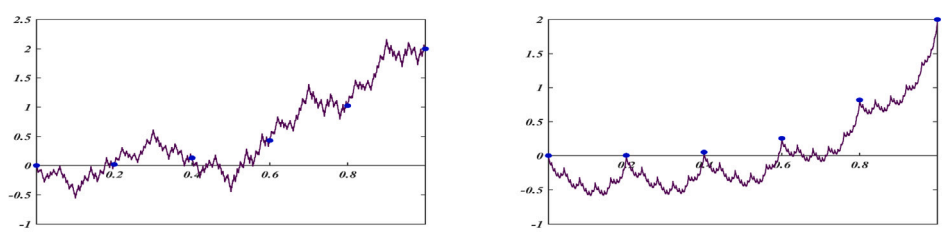


Fig. 7. The fractal function constituted by Fourier series of the set of data points $\{(0,-4),(1/5,-3),(2/5,0),(3/5,3),(4/5,4),(1,-4)\}$ with function scaling factor $c_1(u)=\sin(u), c_2(u)=-0.3u, c_3(u)=0.5-\cos(u), c_4(u)=0.25u, c_5(u)=0.3/e^u$ is generated by $n=1,4$.



(a) α-Fractal function with vertical scaling factor (b) α-FIF with function scaling coefficients

Fig. 8. (a) The related $\mathbf{h}^\alpha$ function is shown on the right side of the visualize with base function $p(u)=2u$ and $\Delta: 0<1/5<2/5<3/5<4/5<1$, and the constant scaling vector, (b) while the same provided data sets with base function and function scaling factors.

Consider the α-Fractal function $\mathbf{h}^\alpha(u)$ with the corresponding data points $\{(u_s,v_s)\in[0,1]\times\mathbb{R} : m=0,1,2,\dots,N\}$. Let $\mathbf{h}^\alpha$ be a periodic function which has a period $2k$ is defined of the form

$$\mathbf{h}^\alpha(u)=\frac{A_0}{2}+\sum_{n=1}^{\infty}\left(A_n\cos\left(\frac{n\pi u}{k}\right)+B_n\sin\left(\frac{n\pi u}{k}\right)\right), \tag{23}$$

which is called the Fourier series of $\mathbf{h}^\alpha(u)$ for all $t\in[-k,k]$

$$A_0=\frac{1}{k}\int_{-k}^{k}\mathbf{h}^\alpha(u)du \tag{24}$$

$$A_n=\frac{1}{k}\int_{-k}^{k}\mathbf{h}^\alpha(u)\cos\left(\frac{n\pi u}{k}\right)du \tag{25}$$

$$B_n=\frac{1}{k}\int_{-k}^{k}\mathbf{h}^\alpha(u)\sin\left(\frac{n\pi u}{k}\right)du, \tag{26}$$

where A_0, A_n and B_n are the Fourier coefficients of $\mathbf{h}^\alpha(u)$ is aforementioned Eqns. from (24) to (26). Suppose $\mathbf{h}^\alpha$ is an even function then the coefficients of B_n is zero, so the Eq. (23) as follows

$$\mathbf{h}^\alpha(u)=\frac{A_0}{2}+\sum_{n=1}^{\infty}A_n\cos\left(\frac{n\pi u}{k}\right) \tag{27}$$

is known as Fourier cosine series of $\mathbf{h}^\alpha$, where A_0 and A_n are mentioned in the Eqs. (24) and (25). If $\mathbf{h}^\alpha$ is an odd function then the coefficients of A_n, $n=0,1,2,\dots$ is zero, so the Eq. (23) turns into

$$\mathbf{h}^\alpha(u)=\sum_{n=1}^{\infty}B_n\sin\left(\frac{n\pi u}{k}\right) \tag{28}$$

called Fourier sine series of $\mathbf{h}^\alpha$, where B_n is mentioned in the Eq. (26).

The following Theorem 4.1 illuminates an even function represented by the Fourier cosine series of $\mathbf{h}^\alpha$ is a α-fractal function on the closed interval $[0,1]$.

Theorem 4.1. *Let $\mathbf{h}^\alpha$ be the α-FIF on the closed interval $[0,1]$. If $\|c_s(u)\|<1$ is the function scaling vector and composed by the transformations is $l_s(u)=\beta_s u+\gamma_s, \mathbf{f}_s(u,v)=c_s(u)v+q_s(u)$, where $q_s(u)=\mathbf{h}(l_s(u))-c_s(u)p_s(u)$, then the Fourier cosine series of $\mathbf{h}^\alpha$ is*

$$\begin{aligned}\mathbf{h}^\alpha(u)=&\frac{2}{K}\sum_{s=1}^{K}\left(\frac{1}{2}c_s(t)\int_0^1\mathbf{h}^\alpha(t)dt-\frac{1}{2}\int_0^1\left(c_s'(t)\int_0^1\mathbf{h}^\alpha(t)dt\right)dt+\frac{1}{2}\int_0^1\mathbf{h}(l_s(t))dt\right.\\&\left.-\frac{1}{2}c_s(t)\int_0^1p_s(t)dt+\frac{1}{2}\int_0^1\left(c_s'(t)\int_0^1p_s(t)dt\right)dt\right)\\&+\frac{2}{K}\sum_{s=1}^{K}\left(\sum_{n=1}^{\infty}\left(\int_0^1\left(c_s(t)\int_0^1\mathbf{h}^\alpha(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt-\int_0^1\left(c_s'(t)\int_0^1\mathbf{h}^\alpha(t)dt\right.\right.\right.\\&\left.\left.\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right)-\left(c_s(t)\int_0^1p_s(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right.\\&\left.\left.+\int_0^1\left(c_s'(t)\int_0^1p_s(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right)+\int_0^1\mathbf{h}(l_s(t))\right.\end{aligned}$$

$$\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\Big)\Big)\cos(n\pi u).$$

Proof. Consider the non affine transformation

$$l_s(u)=\beta_s u+\gamma_s, \mathbf{f}_s(u,v)=c_s(u)v+h(l_s(u))-c_s(u)p_s(u), \quad s=1,2,\dots,K.$$

Since $\mathbf{h}^\alpha$ is a continuous function on $[0,1]$ in (25). If $\mathbf{h}^\alpha$ is an even function then $\mathbf{h}^\alpha$ could be constituted by Fourier cosine series. Then the Fourier coefficients A_n, $\forall n=0,1,2,3,\dots$ can be determined the half range interval $[0,1]$ as follows

$$A_0=\frac{1}{k}\int_{-k}^{k}\mathbf{h}^\alpha(u)du=2\int_0^1\mathbf{h}^\alpha(u)du=2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}}\mathbf{h}^\alpha(u)du.$$

Let the transformation $u=l_s(t)$, where $u=\frac{1}{K}t+\frac{s-1}{K}$, $du=\frac{1}{K}dt$, and substituting the functional equation and utilizing the integration by parts formula so we have

$$A_0=\frac{2}{K}\sum_{s=1}^{K}\Bigg(c_s(t)\int_0^1\mathbf{h}^\alpha(t)dt-\int_0^1\left(c_s'(t)\int_0^1\mathbf{h}^\alpha(t)dt\right)dt \tag{29}$$
$$+\int_0^1\mathbf{h}(l_s(t))dt-\left(c_s(t)\int_0^1 p_s(t)dt+\int_0^1\left(c_s'(t)\int_0^1 p_s(t)dt\right)dt\right)\Bigg).$$

Now find the Fourier coefficient of A_n for $n=1,2,\dots,$

$$A_n=2\int_0^1\mathbf{h}^\alpha(u)\cos(n\pi u)\,du=2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}}\mathbf{h}^\alpha(u)\cos(n\pi u)\,du.$$

Once again by applying the functional equation and the integration by parts procedure to simplify the aforementioned equation as,

$$A_n=\frac{2}{K}\sum_{s=1}^{K}\Bigg(c_s(t)\int_0^1\mathbf{h}^\alpha(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt-\int_0^1\Bigg(c_s'(t)\int_0^1\mathbf{h}^\alpha(t)dt \tag{30}$$
$$\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\Bigg)dt\Bigg)-\frac{2}{K}\sum_{s=1}^{K}\Bigg(c_s(t)\int_0^1 p_s(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt$$
$$+\int_0^1\left(c_s'(t)\int_0^1 p_s(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt+\frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))$$
$$\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\Bigg).$$

As a result, the complete form of the Fourier cosine series of $\mathbf{h}^\alpha$ in the half range interval $[0,1]$ has been mentioned.

Remark 4.1. Theorem 4.1 illustrated the proof of Fourier cosine series depiction of α-Fractal function with function scaling ratio. Suppose the function scaling vector is replaced by the consistent scaling vector in Theorem 4.1, then the following Fourier cosine series α-fractal function is of the form,

$$\mathbf{h}^\alpha(u)=\frac{1}{K}\sum_{s=1}^{K}c_s\int_0^1\left(\mathbf{h}^\alpha(t)-p_s(t)\right)dt+\frac{1}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))dt$$
$$+\sum_{n=1}^{\infty}\left(\frac{2}{K}\sum_{s=1}^{K}c_s\int_0^1\left(\mathbf{h}^\alpha(t)-p_s(t)\right)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right.$$
$$\left.+\frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)\cos(n\pi u).$$

Consider the non affine transformation

$$l_s(u)=\beta_s u+\gamma_s, \mathbf{f}_s(u,v)=c_s u+h(l_s(u))-c_s p_s(u), \quad s=1,2,\dots,K.$$

Since $\mathbf{h}^\alpha$ is a continuous function on $[0,1]$ in (25). If $\mathbf{h}^\alpha$ is an even function then $\mathbf{h}^\alpha$ could be mentioned by Fourier cosine series. The Fourier coefficients A_n determines the half range interval $[0,1]$,

$$A_0=\frac{1}{k}\int_{-k}^{k}\mathbf{h}^\alpha(u)du=2\int_0^1\mathbf{h}^\alpha(u)du=2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}}\mathbf{h}^\alpha(u)dt.$$

Let the transformation $u=l_s(t)$, where $u=\frac{1}{K}t+\frac{s-1}{K}$ and also $dt=Kdu$, so we have

$$A_0=2\sum_{s=1}^{K}\int_0^1\mathbf{h}^\alpha(l_s(t))d\left(\frac{1}{K}t+\frac{s-1}{K}\right)=\frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}^\alpha(l_s(t))dt.$$

By the functional equation $\mathbf{h}^\alpha(l_s(t))=c_s\mathbf{h}^\alpha(t)+q_s(t)$, where $q_s(t)=\mathbf{h}(l_s(t))-c_s p_s(t)$,

$$A_0=\frac{2}{K}\sum_{s=1}^{K}c_s\int_0^1\left(\mathbf{h}^\alpha(t)-p_s(t)\right)dt+\frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))dt.$$

Now determine the Fourier coefficient of A_n for $n=1,2,\dots$ is of the form, and again using the transformation, also applying the functional equation, so obtain the following equation,

$$A_n=\frac{2}{K}\sum_{s=1}^{K}c_s\int_0^1\left(\mathbf{h}^\alpha(t)-p_s(t)\right)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt$$
$$+\frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt.$$

The Fourier cosine series of $\mathbf{h}^\alpha$ of constant scaling factor can be represented by the following

$$\mathbf{h}^\alpha(u)=\frac{1}{K}\sum_{s=1}^{K}c_s\int_0^1\left(\mathbf{h}^\alpha(t)-p_s(t)\right)dt+\frac{1}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))dt \tag{31}$$
$$+\sum_{n=1}^{\infty}\left(\frac{2}{K}\sum_{s=1}^{K}c_s\int_0^1\left(\mathbf{h}^\alpha(t)-p_s(t)\right)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right.$$
$$\left.+\frac{2}{K}\sum_{s=1}^{K}\int_0^1\mathbf{h}(l_s(t))\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)\cos(n\pi u).$$

The aforementioned equation is the complete form of the Fourier cosine series of α-Fractal function with constant scaling vector in the half range interval $[0,1]$.

Example 4.2. Let $\mathbf{h}(u)=2u^4$ and $\Delta: 0<1/5<2/5<3/5<4/5<1$, be the subdivision of a compact interval $[0,1]$ and the function scaling vector are $c_1(u)=-0.3u/6, c_2(u)=0.5-\sin(10u), c_3(u)=-cos(u/15), c_4(u)=(u+1)/5, c_5(u)=0.5u$. Let $p_s:[0,1]\to\mathbb{R}$ be a continuous function on $[0,1]$. So now choose the base functions $p_1(u)=2u, p_2(u)=2u^2, p_3(u)=2(\sin(u)/\sin(1)), p_4(u)=2(e^u-1/e-1)$ which gratifies the join-up prerequisites $h(0)=p_s(0)$ and $h(1)=p_s(1)$ but $\mathbf{h}(u)\neq p_s(u)$. The Fourier cosine series of α-fractal function is portrayed in Fig. 10.

The visualization of the α-FIF with a variable scaling factor, shown in Fig. 10, is smoother than the constant scaling factor, revealed in Fig. 9. Fig. 10 presents that the fractal function irregularities reduce with increasing harmonic iteration as opposed in Fig. 9.

Theorem 4.2. *Let $\mathbf{h}^\alpha$ be the α-Fractal function on the closed interval $[0,1]$. If $\|c_s(u)\|<1$ is the function scaling factor and composed by the transformations $l_s(u)=\beta_s u+\gamma_s, \mathbf{f}_s(u,v)=c_s(u)v+q_s(u)$, where $q_s(u)=\mathbf{h}(l_s(u))-c_s(u)p_s(u)$. Then the Fourier sine series of $\mathbf{h}^\alpha$ is given by*

$$\mathbf{h}^\alpha(u)=\frac{2}{K}\sum_{s=1}^{K}\Bigg(\sum_{n=1}^{\infty}\Bigg(\Bigg(c_s(t)\int_0^1\mathbf{h}^\alpha(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt-\int_0^1\Bigg(c_s'(t)\int_0^1$$
$$\mathbf{h}^\alpha(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\Bigg)dt\Bigg)+\Bigg(c_s(t)\int_0^1 p_s(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt$$
$$-\int_0^1\left(c_s'(t)\int_0^1 p_s(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\Bigg)+\int_0^1\mathbf{h}(l_s(t))$$
$$\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\Bigg)\Bigg)\sin(n\pi u).$$

Proof. Consider the non affine transformation

$$l_s(u)=\beta_s u+\gamma_s, \mathbf{f}_s(u,v)=c_s(u)v+h(l_s(u))-c_s(u)p_s(u), \ s=1,2,\dots,K.$$

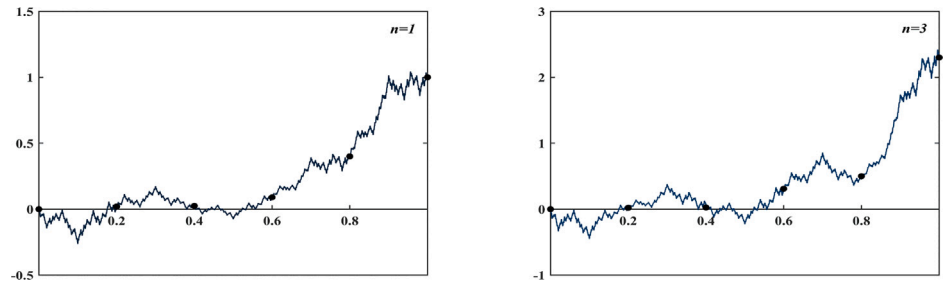


Fig. 9. The FIF constituted by Fourier cosine series for the provided data points $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, with constant scaling factors $c_1 = 0.4, c_2 = -0.4, c_3 = 0.5, c_4 = -0.5, c_5 = -0.5$, and $p_1(u) = 2u, p_2(u) = 2u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_4(u) = 2(e^u - 1/e - 1)$, this meets the prerequisites for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$, is generated by $n = 1, 3$.

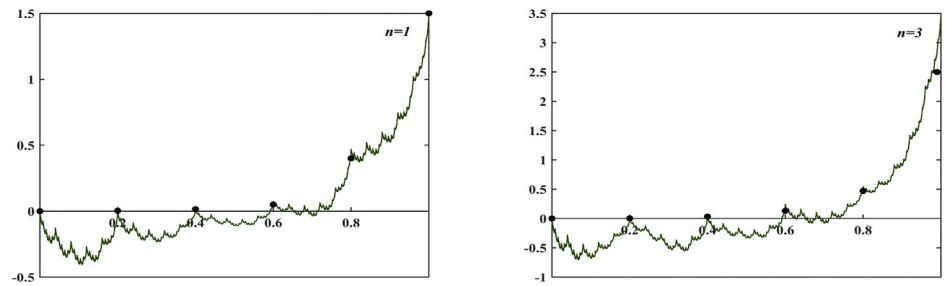


Fig. 10. The fractal interpolation function constituted by Fourier cosine series for the provided data points $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, with function scaling factors $c_1(u) = -0.3u/6, c_2(u) = 0.5 - \sin(10u), c_3(u) = -cos(u/15), c_4(u) = (u+1)/5, c_5(u) = 0.5u$, and $p_1(u) = 2u, p_2(u) = 2u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_4(u) = 2(e^u - 1/e - 1)$, this meets the prerequisites for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$, is generated by $n = 1, 3$.

Since $\mathbf{h}^\alpha$ is a continuous function on $[0, 1]$ in (25). If $\mathbf{h}^\alpha$ is an odd function, then $\mathbf{h}^\alpha$ could be referred to by Fourier sine series. The Fourier coefficients $B_n, \forall n = 1, 2, 3, \ldots$ can be determined the half range interval $[0, 1]$ as follows

$$B_n = 2\int_0^1 \mathbf{h}^\alpha(u)\sin(n\pi u)\,du = 2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}} \mathbf{h}^\alpha(u)\sin(n\pi u)\,du$$

Consider the transformation $u = l_s(t)$, where $u = \frac{1}{K}t + \frac{s-1}{K}$, $dt = Kdu$, also applying the functional equation $\mathbf{h}^\alpha(l_s(t)) = c_s(t)\mathbf{h}^\alpha(t) + q_s(t)$. So, we obtain the following equation

$$B_n = \frac{2}{K}\sum_{s=1}^{K}\int_0^1 c_s(t)\mathbf{h}^\alpha(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt + \frac{2}{K}\sum_{s=1}^{K}\int_0^1 \left(\mathbf{h}(l_s(t)) - c_s(t)p_s(t)\right)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt.$$

Using the integration by parts formula to simplify the equation as stated above, one can be obtained,

$$B_n = \frac{2}{K}\sum_{s=1}^{K}\left(c_s(t)\int_0^1 \mathbf{h}^\alpha(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\left(c_s'(t)\int_0^1 \mathbf{h}^\alpha(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right) + \frac{2}{K}\sum_{s=1}^{K}\left(c_s(t)\int_0^1 p_s(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\left(c_s'(t)\int_0^1 p_s(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right) + \frac{2}{K}\sum_{s=1}^{K}\int_0^1 \mathbf{h}(l_s(t))\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt. \tag{32}$$

As a result, the complete form of the Fourier sine series of $\mathbf{h}^\alpha$ with function scaling vector in the half range interval $[0, 1]$ has been mentioned.

Remark 4.2. Consider the non affine transformation of the form is

$$l_s(u) = \frac{1}{K}u + \frac{s-1}{K}, \mathbf{f}_s(u, v) = c_s v + h(l_s(u)) - c_s p_s(u), \ s = 1, 2, \ldots, K.$$

Since $\mathbf{h}^\alpha$ is a continuous function on $[0, 1]$ in (25). If $\mathbf{h}^\alpha$ is an odd function, then $\mathbf{h}^\alpha$ is expressed by Fourier sine series. The Fourier coefficients B_n on the half range interval $[0, 1]$, then

$$B_n = 2\int_0^1 \mathbf{h}^\alpha(u)\sin(n\pi u)\,du = 2\sum_{s=1}^{K}\int_{\frac{s-1}{K}}^{\frac{s}{K}} \mathbf{h}^\alpha(u)\sin(n\pi u)\,du.$$

Let the transformation $u = l_s(t)$, where $u = \frac{1}{K}t + \frac{s-1}{K}$, $dt = Kdu$, and using the functional equation is $\mathbf{h}^\alpha(l_s(t)) = c_s\mathbf{h}^\alpha(t) + q_s(t)$, where $q_s(t) = \mathbf{h}(l_s(t)) - c_s p_s(t)$, so obtained the following equation,

$$B_n = \frac{2}{K}\sum_{s=1}^{K} c_s\int_0^1 \left(\mathbf{h}^\alpha(t) - p_s(t)\right)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt + \frac{2}{K}\sum_{s=1}^{K}\int_0^1 \mathbf{h}(l_s(t))\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt.$$

The Fourier sine series of $\mathbf{h}^\alpha$ can be expressed as follows

$$\mathbf{h}^\alpha(u) = \sum_{n=1}^{\infty}\left(\frac{2}{K}\sum_{s=1}^{K} c_s\int_0^1 \left(\mathbf{h}^\alpha(t) - p_s(t)\right)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt + \frac{2}{K}\sum_{s=1}^{K}\int_0^1 \mathbf{h}(l_s(t))\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)\sin(n\pi u). \tag{33}$$

This is a complete demonstration that the Fourier sine series is composed of α-Fractal function with constant scaling factor. That is, in Theorem 4, if the function scaling vector are substituted by constant scaling vector provides the proof of Remark 4.

Example 4.3. Let $\mathbf{h}(u) = 2u^3$ and $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, be the subdivision of a compact interval $[0, 1]$ and the function scaling vector are $c_1(u) = -0.3u/6, c_2(u) = 0.5 - \sin(10u), c_3(u) = -cos(u/15), c_4(u) = (u+1)/5, c_5(u) = 0.5u$. Let p_s be a continuous mapping on $[0, 1]$. So now choose the base functions $p_1(u) = 2u, p_1(u) = u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_3(u) = 2(e^u - 1/e - 1)$ this meets the conditions for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$. The Fourier sine series of α-fractal function is depicted in Fig. 12.

The shape of the α-fractal function with a variable scaling factor displayed in Fig. 12, is more elegant than the curve of the constant

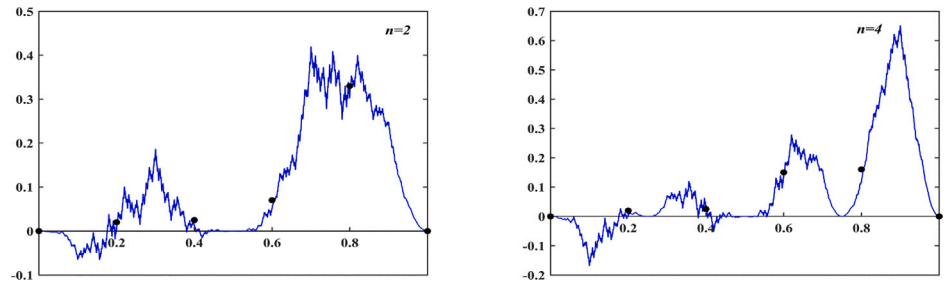


Fig. 11. The fractal function constituted by Fourier sine series for the provided data points $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, with constant scaling factors $c_1 = 0.4, c_2 = -0.4, c_3 = 0.5, c_4 = -0.5, c_5 = -0.5$, and $p_1(u) = 2u, p_2(u) = 2u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_4(u) = 2(e^u - 1/e - 1)$, this meets the conditions for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$, is generated by $n = 2, 4$.

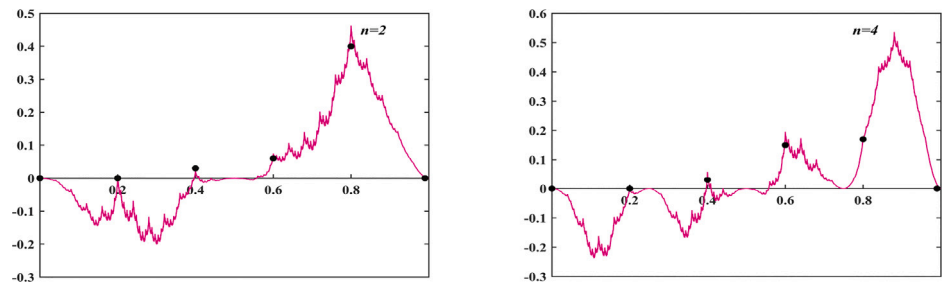


Fig. 12. The fractal interpolation function constituted by Fourier sine series for the provided data points $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, with function scaling factors $c_1(u) = 0.3u, c_2(u) = 0.5\sin(u), c_3(u) = 0.5\cos(u), c_4(u) = (u+1)/5$, and $p_1(u) = 2u, p_2(u) = 2u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_4(u) = 2(e^u - 1/e - 1)$, this meets the conditions for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$, is generated by $n = 2, 4$.

scaling factor, shown in Fig. 11. For $n = 2, 4$, the intricacies of the α-fractal function in Fig. 12 are much less severe than those in Fig. 11. In the same manner, Figs. 13 and 14 revealing the fractal function of the Fourier series on the closed interval $[0, 1]$. The fluctuations in Figs. 13 and 14 are identical to those in Figs. 8(a) and 8(b), but the ranges are diverse. The Next Corollary 4.1 demonstrates the function represented by the Fourier series of $\mathbf{h}^\alpha$ is a Fourier α-Fractal function on the bounded interval $[0, 1]$.

Corollary 4.1. *Suppose $\mathbf{h}^\alpha$ is the α-FIF on the closed interval $[0, 1]$ generated by the non affine transformation form is $l_s(u) = \beta_s u + \gamma_s, \mathbf{f}_s(u, v) = c_s v + q_s(u)$, where $q_s(u) = \mathbf{h}(l_s(u)) - c_s p_s(u)$. Then the Fourier series of $\mathbf{h}^\alpha$ is expressed as follows*

$$\begin{aligned}\mathbf{h}^\alpha(u) = {} & \frac{1}{K}\sum_{s=1}^{K} c_s \int_0^1 \left(\mathbf{h}^\alpha(t) - p_s(t)\right) dt + \frac{1}{K}\sum_{i=1}^{K}\int_0^1 \mathbf{h}(l_s(t))dt \\ & + \sum_{n=1}^{\infty}\left(\frac{2}{K}\sum_{s=1}^{K} c_s \int_0^1 \left(\mathbf{h}^\alpha(t) - p_s(t)\right)\cos\left(\frac{n\pi}{K}(t+s-1)\right) dt\right. \\ & \left. + \frac{2}{K}\sum_{i=1}^{K}\int_0^1 \mathbf{h}(l_s(t))\cos\left(\frac{n\pi}{K}(t+s-1)\right) dt\right)\cos(n\pi u) \\ & + \sum_{n=1}^{\infty}\left(\frac{2}{K}\sum_{s=1}^{K} c_s \int_0^1 \left(\mathbf{h}^\alpha(t) - p_s(t)\right)\sin\left(\frac{n\pi}{K}(t+s-1)\right) dt\right. \\ & \left. + \frac{2}{K}\sum_{s=1}^{K}\int_0^1 \mathbf{h}(l_s(t))\sin\left(\frac{n\pi}{K}(t+s-1)\right) dt\right)\sin(n\pi u).\end{aligned} \tag{34}$$

Proof. The domain of definition of $\mathbf{h}^\alpha$ is extended to an all-inclusive collection of real numbers $\mathbb{R}$, when the α-Fractal function $\mathbf{h}^\alpha$ is continuous on the interval $[0, 1]$. Even though know that Eq. (31) plus Eq. (33) equals Eq. (34), the Fourier series representation in Eq. (23). By applying the conclusions from Remarks 4.1 and 4.2 completes Corollary 4.1 proof.

Corollary 4.2. *Let $\mathbf{h}^\alpha$ be the α-fractal function on the closed interval $[0, 1]$. If $\|c_s(u)\| < 1$ is the function scaling factor and compose by the non affine transformation is $l_s(u) = \beta_s u + \gamma_s, \mathbf{f}_s(u, v) = c_s(u)v + q_s(u)$, where $q_s(u) = \mathbf{h}(l_s(u)) - c_s(u)p_s(u)$. Then the Fourier series of $\mathbf{h}^\alpha$ could be expressed by the follows as*

$$\begin{aligned}\mathbf{h}^\alpha(u) = {} & \frac{2}{K}\sum_{s=1}^{K}\left(\frac{1}{2}c_s(t)\int_0^1 \mathbf{h}^\alpha(t)dt - \frac{1}{2}\int_0^1\left(c_s'(t)\int_0^1 \mathbf{h}^\alpha(t)dt\right)dt\right. \\ & \left. + \frac{1}{2}\int_0^1 \mathbf{h}(l_s(t))du - \frac{1}{2}c_s(t)\int_0^1 p_s(t)dt + \frac{1}{2}\int_0^1\left(c_s'(t)\int_0^1 p_s(t)dt\right)dt\right) \\ & + \frac{2}{K}\sum_{s=1}^{K}\left(\sum_{n=1}^{\infty}\left(\left(c_s(t)\int_0^1 \mathbf{h}^\alpha(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\left(c_s'(t)\int_0^1 \mathbf{h}^\alpha(t)dt\right.\right.\right.\right. \\ & \left.\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right) + \left(c_s(t)\int_0^1 p_s(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right. \\ & \left. - \int_0^1\left(c_s'(t)\int_0^1 p_s(t)\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right) + \int_0^1 \mathbf{h}(l_s(t)) \\ & \left.\cos\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)\right)\cos(n\pi u) + \frac{2}{K}\sum_{s=1}^{K}\left(\sum_{n=1}^{\infty}\left(\left(c_s(t)\int_0^1 \mathbf{h}^\alpha(t)\right.\right.\right. \\ & \left.\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\left(c_s'(t)\int_0^1 \mathbf{h}^\alpha(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right) \\ & + \left(c_s(t)\int_0^1 p_s(t)\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt - \int_0^1\left(c_s'(t)\int_0^1 p_s(t)\right.\right. \\ & \left.\left.\left.\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)dt\right) + \int_0^1 \mathbf{h}(l_s(t))\sin\left(\frac{n\pi}{K}(t+s-1)\right)dt\right)\right)\sin(n\pi u).\end{aligned} \tag{35}$$

Proof. The $\mathbf{h}^\alpha$ is extended to an all-inclusive set of real numbers $\mathbb{R}$ when the α-Fractal function $\mathbf{h}^\alpha$ is continuous on the interval $[0, 1]$. Even though know that Eq. (30) plus Eq. (32) equals Eq. (35), the Fourier series representation in Eq. (23). By applying the conclusions from Theorems 4.1 and 4.2 completes Corollary 4.2 proof.

Remark 4.3. When $\mathbf{h}(u)$ is an even or odd function, then $\mathbf{h}(l_s(t))$ does not need to be an even or odd function. Non-affine fractal functions approximate the any functions on a compact interval I and is defined based on $q_s(u)$. In this paper, we choose the non-affine case as $q_s(u) = \mathbf{h}(l_s(u)) - c_s(u)p_s(u)$. This function depends on both $p_s(u)$ and $\mathbf{h}(l_s(u))$ with scaling factor $\|c_s(u)\| \leq 1$. The function $p_s(u)$ can either an even

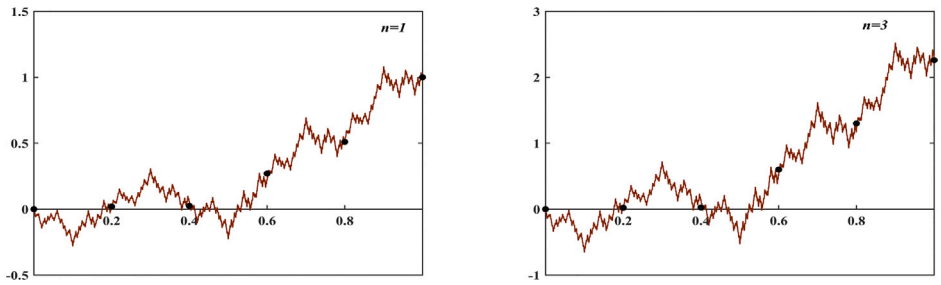


Fig. 13. The fractal interpolation function constituted by Fourier series for the provided data points $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, with constant scaling factors $c_1 = 0.4, c_2 = -0.4, c_3 = 0.5, c_4 = -0.5, c_5 = -0.5$, and $p_1(u) = 2u, p_2(u) = 2u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_4(u) = 2(e^u - 1/e - 1)$,this meets the conditions for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$, is generated by $n = 1, 3$.

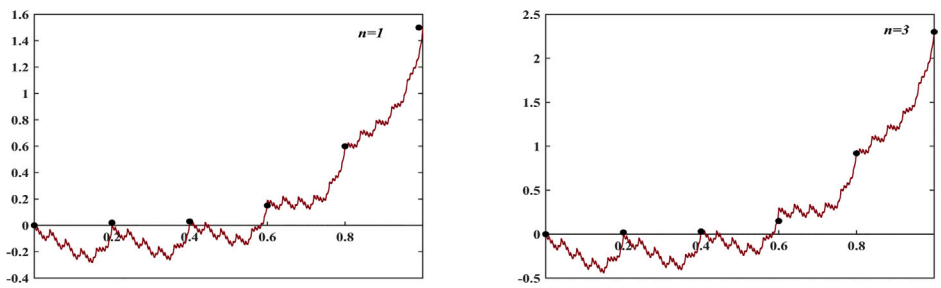


Fig. 14. The fractal interpolation function constituted by Fourier series for the provided data points $\Delta : 0 < 1/5 < 2/5 < 3/5 < 4/5 < 1$, with function scaling factors $c_1(u) = -0.3u/6, c_2(u) = 0.5 - \sin(10u), c_3(u) = -cos(u/15), c_4(u) = (u+1)/5, c_5(u) = 0.5u$, and $p_1(u) = 2u, p_2(u) = 2u^2, p_3(u) = 2(\sin(u)/\sin(1)), p_4(u) = 2(e^u - 1/e - 1)$,this meets the conditions for joining $h(0) = p_s(0)$ and $h(1) = p_s(1)$ but $\mathbf{h}(u) \neq p_s(u)$, where $s = 1, 2, 3, 4$, is generated by $n = 1, 3$.

or odd function, while $\mathbf{h}(l_s(u))$ is the composition function of $l_s(u) = 1/Ku + (s-1)/K$. As a result, $q_s(u)$ is neither an odd nor even function.

5. Conclusion

This article demonstrated how to construct Fourier cosine and Fourier sine series for various fractal functions, including affine and non-affine functions with constant and variable scaling vectors. Additionally, instead of a function scaling vector, consistent factors were chosen and investigated for both affine fractal functions and α-fractal functions. The given concepts have been illustrated with appropriate examples and graphical approaches. Because non-smoothness is essential for approximating the complexity of irregular signals, the Fourier series associated with fractal functions offers the closest approach to indicating periodic signals.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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